Question

Adamantane frameworks are not only found in organic compounds, but also commonly seen in inorganic clusters, of which P_4O_{10} is a typical representative.

1. Dissolve $GeBr_4$ in CS_2 containing Al_2Br_6, and then introduce H_2S gas into it to obtain a similar compound $\mathbf{A}$, containing $\omega_{Ge} = 0.3620$.

2. The structure of P_4O_{10} can be described as $P_4 \cdot O_6 @ O_4$, that is, a set of interpenetrating P_4 polyhedra and O_6 polyhedra encapsulated in O_4 polyhedra. Try to describe the structure of adamantane (substance $\mathbf{B}$) in this way, and give the corresponding polyhedra for each layer.

3. Adding more atoms to the periphery of the adamantane framework can result in more complex cluster structures. There is a highly symmetrical anion $[Mo_mS_xCu_nCl_y]^{4-}$ (substance $\mathbf{C}$), containing $\omega_{Cu} = 0.4945$, $\omega_{Mo} = 0.0747$, and each element has a common valence state. Where Cl is 2-coordinate and has only one chemical environment, and S is 4-coordinate.

Which of the following statements is correct?

A. The structure of $\mathbf{A}$ is $Ge_4S_4Br_6$

B. $\mathbf{A}$ contains the element Al

C. In $\mathbf{A}$, a Ge atom connects to two S atoms and a Br atom.

D. Following the notation of $P_4 \cdot O_6 @ O_4$, substance $\mathbf{B}$ can be written as $C_4 \cdot C_6 @ H_8 H_8$, and the polyhedra from left to right are a regular tetrahedron, a regular octahedron, a regular hexahedron, and a regular hexahedron.

E. Following the notation of $P_4 \cdot O_6 @ O_4$, the substance $\mathbf{B}$ can be written as $C_4 \cdot C_6 @ H_{12} H_4$, the polyhedra from left to right are regular tetrahedron, regular octahedron, regular icosahedron, regular tetrahedron.

F. In substance $\mathbf{C}$, $m=1, n=10, x=2, y=14$

G. Substance $\mathbf{C}$ with $m=1, n=10, x=3, y=13$

H. In substance $\mathbf{C}$, $m=1, n=10, x=6, y=10$.

I. Each Cu atom in substance $\mathbf{C}$ is connected to S.

J. Substance $\mathbf{C}$ is divided into core and periphery, the core being $\mathrm{Mo@S_4{\cdot}Cu_6}$, and the periphery being $\mathrm{Cu_4{\cdot}Cl_{12}}$

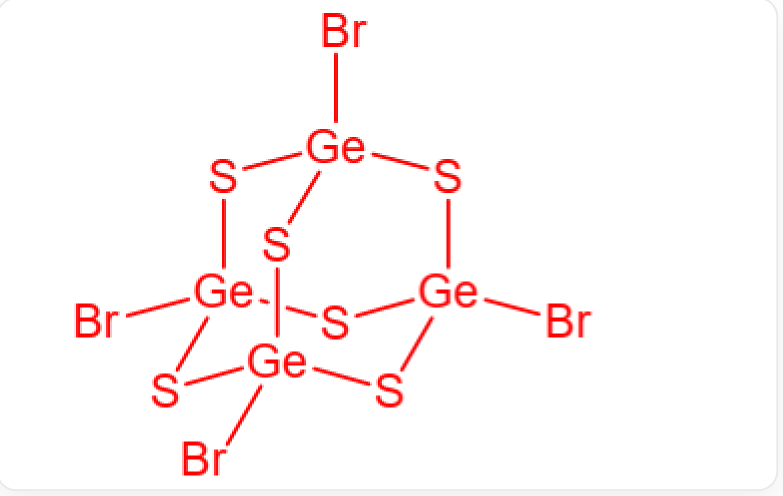
K. All of the above options are incorrect.

Answer

Correct Answer: J

Detailed Explanation

First question



Structure diagram of substance A, composed of Ge4S6Br4, where Ge is connected to 3 S and 1 Br

According to the question, the structure of substance $\mathbf{A}$ is consistent with that of P_4O_{10}. The possible constituent elements of this substance are Ge ˊS ˊBr, where Ge is most likely to replace the position of P, S should be 2-coordinated and located on the bridge, and Br is more suitable as a terminal group. The resulting substance has the composition $Ge_4S_6Br_4$.

Verification of ω_{Ge}: The relative molecular mass of the chemical formula is $4{\times}72.63 + 6{\times}32.07 + 4{\times}79.90 = 290.52 + 192.42 + 319.60 = 802.54$

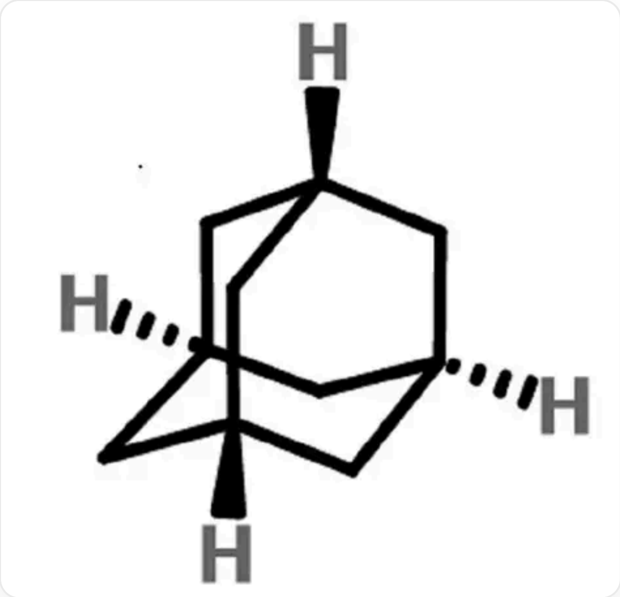
Then $\omega_{Ge} = (4{\times}72.63)/802.54 = 290.52/802.54 \approx 0.3620$, which meets the requirements of the question.

Therefore, the composition of substance $\mathbf{A}$ is $Ge_4S_6Br_4$, where Ge is connected to 3 S and 1 Br, ABC are incorrect.

CHECKPOINT 1 PTS

The composition of substance $\mathbf{A}$ is $Ge_4S_6Br_4$, where Ge is connected to 3 S and 1 Br, ABC are incorrect

Second question



Structure diagram of adamantane, which can be described as C4·C6@H12·H4, from left to right are regular tetrahedron, regular octahedron, truncated regular tetrahedron, regular tetrahedron

According to the writing method of $P_4{\cdot}O_6@O_4$, substance $\mathbf{B}$ can be written as $C_4{\cdot}C_6@H_{12}H_4$. The polyhedra from left to right are regular tetrahedron, regular octahedron, truncated regular tetrahedron, regular tetrahedron, DE are incorrect.

CHECKPOINT 1 PTS

According to the writing method of $P_4 \cdot O_6 @ O_4$, substance $\mathbf{B}$ can be written as $C_4 \cdot C_6 @ H_{12} H_4$. The polyhedra from left to right are regular tetrahedron, regular octahedron, truncated regular tetrahedron, regular tetrahedron, DE are incorrect

Third question

$$\mathrm{Cu} : \mathrm{Mo} = (0.4945/63.55) : (0.0747/95.96) = 10 : 1$$

CHECKPOINT 1 PTS

$$\mathrm{Cu} : \mathrm{Mo} = 10 : 1$$

Assuming that it contains only 1 Mo, i.e., $m = 1$, then the sum of the molar masses of S and Cl in this ion is:

$$10 \times 63.55 \times (1 - 0.4945)/0.4945 - 95.96 = 553.7$$

CHECKPOINT 1 PTS

The sum of the molar masses of S and Cl is 553.7

$$32.07x + 35.45y = 553.7$$

The integer solution can be obtained by listing: $x = 4, y = 12$

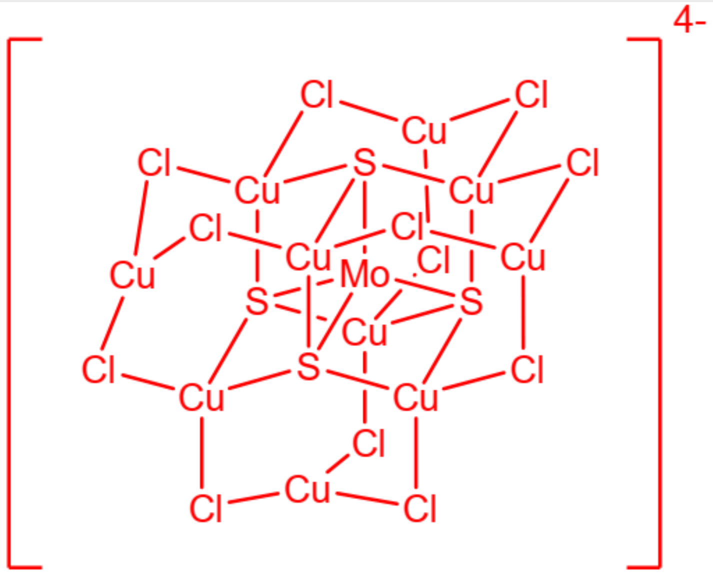
At this time, Cl is -1 valence, S is -2 valence, Cu is $+1$ valence, and Mo is $+6$ valence, which are in line with common valence states.

If $m > 1$, then 0 valence Cu or non-$+6$ valence Mo will appear, so it is discarded.

Therefore, $m = 1$, $n = 10$, $x = 4$, $y = 12$, FGH are incorrect.

CHECKPOINT 1 PTS

$m = 1$, $n = 10$, $x = 4$, $y = 12$, FGH are incorrect



Structure diagram of MoCu10S4Cl12, where the core is Mo@S4·Cu6, and the periphery is Cu4·Cl12

Substance **C** is $\mathrm{MoCu_{10}S_4Cl_{12}}$, where the core is $\mathrm{Mo@S_4{\cdot}Cu_6}$, and the periphery is $\mathrm{Cu_4{\cdot}Cl_{12}}$. The peripheral Cu is only connected to Cl, I is incorrect, and J is correct.

CHECKPOINT 1 PTS

Substance **C** is $\mathrm{MoCu_{10}S_4Cl_{12}}$, where the core is $\mathrm{Mo@S_4{\cdot}Cu_6}$, and the periphery is $\mathrm{Cu_4{\cdot}Cl_{12}}$, I is incorrect, and J is correct